Week 1

Problem 1: Finding Complexity using Counter Method

```
#include<stdio.h>
int counter=0;
int main()
{
    int i=1,n;
    counter++;
    int s=1;
    counter++;
    scanf("%d",&n);
    while(s<=n)
    {
        counter++;
        i++;
        counter++;
        s+=i;
        counter++;
    }
    counter++;
    printf("%d",counter);
}
```

Problem 2: Finding Complexity using Counter Method

```
#include<stdio.h>
int count=0;
int main(){
    int n;
    scanf("%d",&n);
    count++;
    if(n==1){
        //printf("*");
    }
    else{
        for(int i=1;i<=n;i++){
            count++;
            for(int j=1;j<=n;j++){
                count++;
                count++;
                //printf("*");
                count++;
                //printf("*");
                break;
            }
            count++;
        }
        count++;
    }
    printf("%d",count);
}
```

Problem 3: Finding Complexity using Counter Method

```
#include<stdio.h>
int count=0;
int factor(int num){
    for(int i=1;i<=num;++i){
        count++;
        count++;
        if(num%i==0){
            count++;
            //printf("%d ",i);
        }
    }
    count++;
    return 0;
}
int main(){
    int num;
    scanf("%d",&num);
    factor(num);
    printf("%d",count);
```

```
}
```

Problem 4: Finding Complexity using Counter Method

```
#include<stdio.h>
void function(int n){
    int count=0;
    count++;
    for(int i=n/2;i<n;i++){
        count++;
        for(int j=1;j<n;j=2*j){
            count++;
            for(int k=1;k<n;k=k*2){
                count++;
                count++;
            }
            count++;
        }
        count++;
    }
    count++;
    printf("%d",count);
}
int main(){
    int n;
    scanf("%d",&n);
    function(n);
}
```

Problem 5: Finding Complexity using Counter method

```
#include<stdio.h>
int count=0;
void reverse(int n){
    int rev=0,remainder;
    count++;
    while(n!=0){
        count++;
        remainder=n%10;
        count++;
        rev=rev*10+remainder;
        count++;
        n/=10;
        count++;
    }
    count++;
    count++;
    //printf(rev)
    printf("%d",count);
}
int main(){
    int n;
    scanf("%d",&n);
    reverse(n);
}
```